

LARGE-SCALE VIDEO-BASED SPORTS ACTION CLASSIFICATION USING A HYBRID CNN-LSTM ARCHITECTURE ON THE UCF101 DATASET

Course Project for AIN3002 Deep Learning

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Introduction

Video Action Recognition (VAR) is a core challenge in computer vision that requires modeling both spatial visual patterns and temporal sequences. This project focuses on classifying human actions in videos using a hybrid CNN-LSTM architecture. Unlike traditional methods, our model is trained entirely from scratch on the UCF101 dataset, targeting 10 popular sports categories to evaluate the synergy between convolutional feature extraction and recurrent temporal modeling.

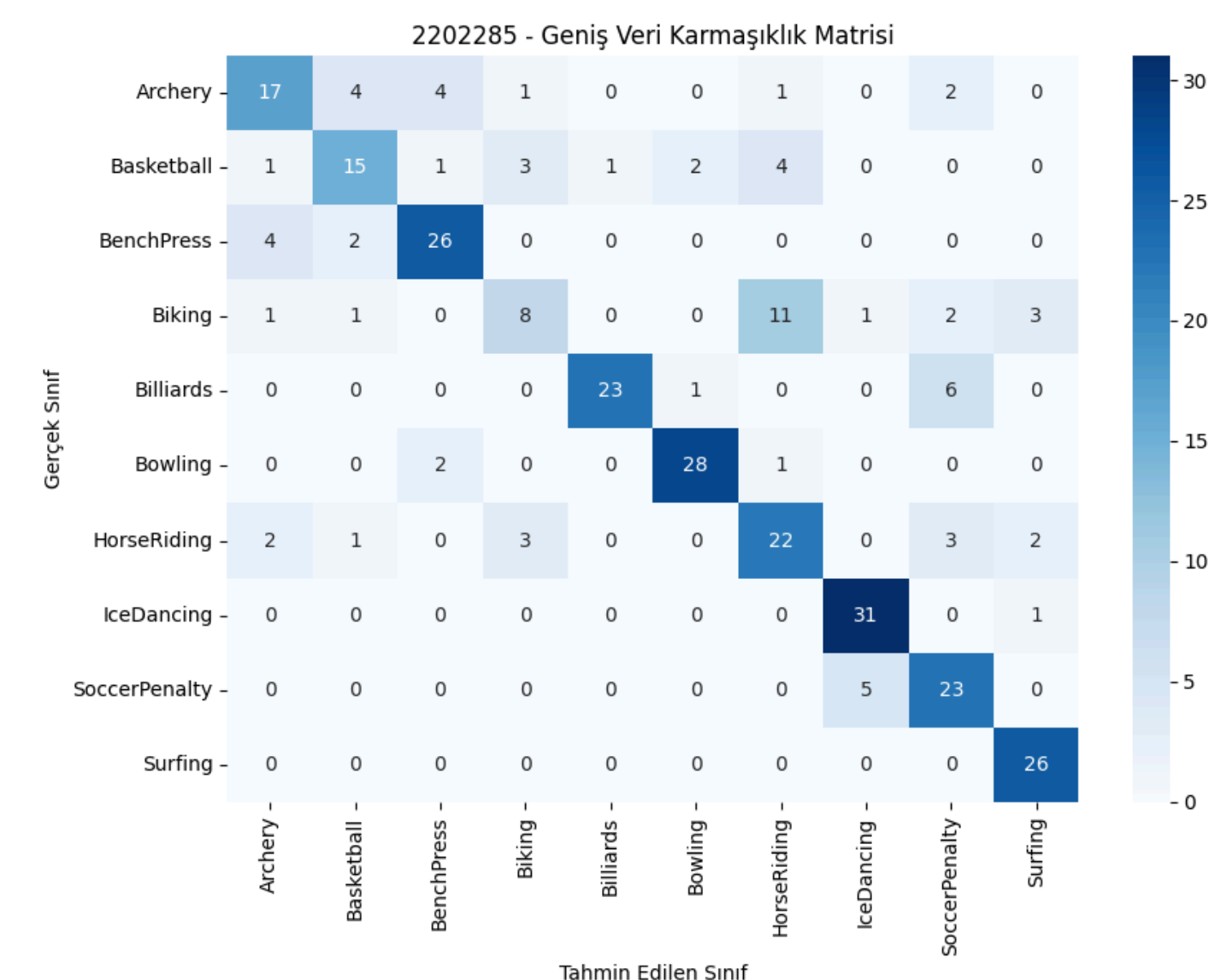


Fig 2: Confusion Matrix

Objective

- Temporal Modeling: Capture the dynamic flow of sports actions across 16 sampled frames per video.
- Memory Optimization: Balance batch size and sequence length to prevent CUDA Out-of-Memory (OOM) errors during training.
- Baseline Comparison: Outperform the theoretical random guessing accuracy (10% for 10 classes) by a significant margin using a non-pretrained deep architecture.
- Performance Target: Achieve over 70% validation accuracy through dynamic data augmentation and Dropout regulation.

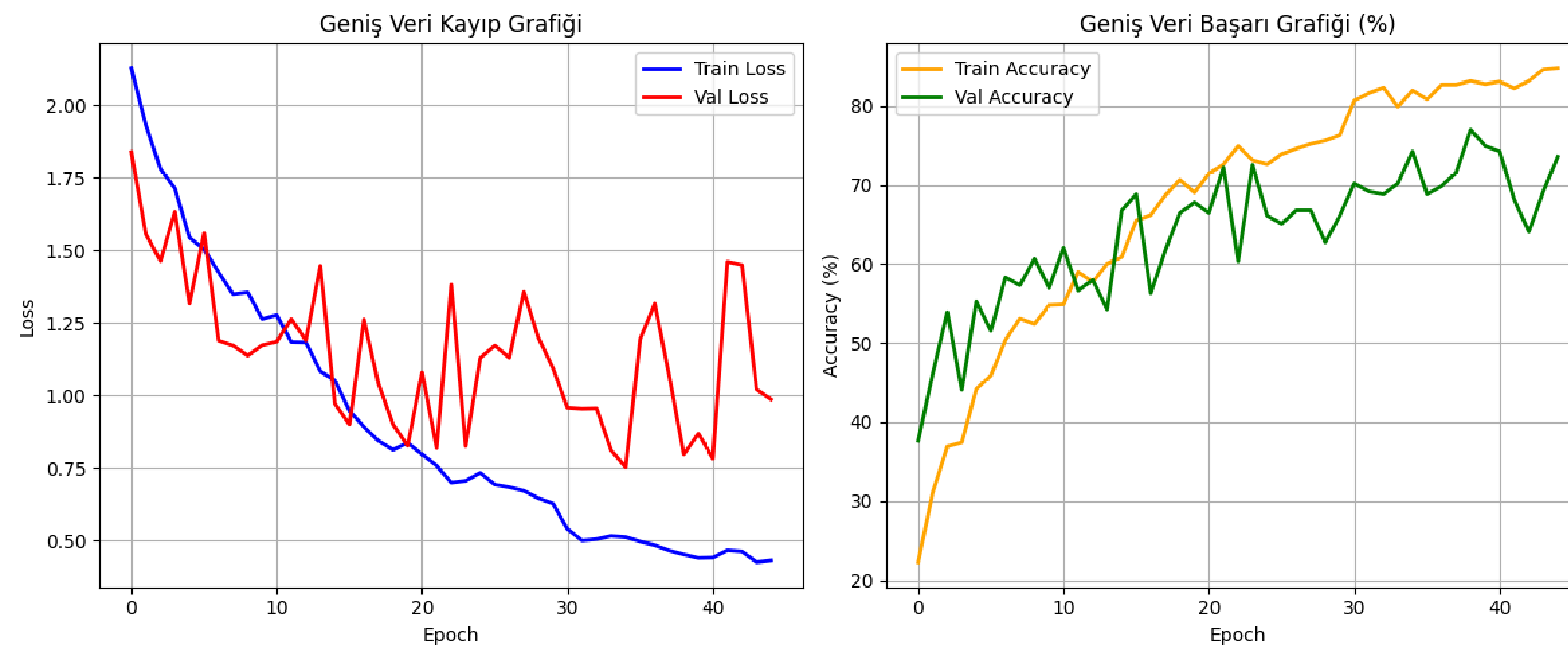


Fig 1: Training Loss & Accuracy

Methodology

- 1** Frame Sampling: Using OpenCV to dynamically extract exactly 16 uniform RGB frames (224 X 224 X 3) from each raw video to stabilize memory footprint.
- 2** Spatial CNN Core: Passing each frame through 4 sequential Convolutional layers (3 X 3 filters) with increasing channel depths: 32, 64, 128, and 256.
- 3** Temporal LSTM Core: Feeding the 256-dimensional spatial sequence vectors into a 2-layer Long Short-Term Memory (LSTM) network with 256 hidden units.
- 4** Regularization & Pooling: Integrating Batch Normalization and ReLU after each Conv layer, finalized by a 2x2 Max Pooling and an Adaptive Average Pooling.
- 5** Dropout & Classification: Applying a 30% Dropout rate between LSTM layers to prevent overfitting, averaging temporal outputs, and classifying via a 10-class Linear layer.

Results

The model was trained for 45 Epochs using the Adam optimizer with a learning rate of 0.0005. Final metrics achieved:

- Training Accuracy: 84.76%
- Validation Accuracy: 73.56%
- Theoretical Baseline: Outperformed random guessing (10%) by over 7.3x.

The ~11% gap between training and validation accuracy indicates a healthy convergence, proving that the model successfully generalized from the UCF101 dataset without significant overfitting.

Conclusion

This study confirms that a hybrid CNN-LSTM architecture, even when trained from scratch, can effectively model complex sports actions by capturing both spatial features and temporal dependencies. The achieved 73.56% accuracy demonstrates robust performance under constrained computational resources. Future Work: Future research will explore integrating 3D-Convolutional layers to further enhance spatiotemporal feature representation and scaling the architecture to the full 101-class UCF dataset.

Digital Project Artifacts



Key Sources & Acknowledgements

Key Sources: [1] Soomro, K. et al. "UCF101: A dataset of 101 human actions classes from videos in the wild." arXiv:1212.0402, 2012. [2] Paszke, A. et al. "PyTorch: An imperative style, high-performance deep learning library." NeurIPS, 2019.

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